

1 / 20 | Ages 3-4 | 10 minutes

Hand shadow shapes

What we are practising

Observe how an object blocks light to make a shadow.

What you need

Torch / Plain wall

Use daylight shadows on the ground; never look at the sun.

Before you begin

An adult points the torch at the wall in a dim, clear space.

An adult holds the torch; never shine it into eyes. Keep the floor clear.

Try it together

1. Put a hand between the torch and wall and find its shadow.
2. Move the hand closer to the torch while the torch stays still.
3. Return it to the starting place and compare the shadow sizes.

Talk, draw or show your idea

What happened when your hand moved towards the light?

Space to explore

2 / 20 | Ages 3-4 | 15 minutes

Float or sink tray

What we are practising

Make a prediction and compare it with an observation.

What you need

Wide shallow tray / Water / Large wooden spoon / Large metal spoon / Unbreakable empty cup / Towel

Use two different large bath toys if spoons are unavailable.

Before you begin

An adult puts the tray on a towel and adds only enough water for the test objects to float freely without touching the bottom.

An adult stays within reach throughout, prevents drinking, wipes spills and empties the tray immediately afterwards.

Try it together

1. Predict what the wooden spoon will do, then lower it gently.
2. Try the metal spoon and describe where each rests.
3. Try the empty cup upright, then tip water into it and observe any change.

Talk, draw or show your idea

Did everything behave as you expected?

Space to explore

3 / 20 | Ages 3-4 | 10 minutes

Move it with air

What we are practising

Notice that moving air can push light objects.

What you need

Two full sheets of paper

An adult can fan with a thin book.

Before you begin

Fold one sheet into a simple fan; loosely fold the other into a standing tent.

Stay with your child and use large, intact, age-appropriate materials.

Try it together

1. Set the paper tent on a smooth table.
2. Wave the fan towards it without touching it.
3. Try from nearer and farther away and compare the movement.

Talk, draw or show your idea

What pushed the paper even though the fan did not touch it?

Space to explore

4 / 20 | Ages 3-4 | 15 minutes

A gentle ramp test

What we are practising

Compare a change while keeping the toy and release point the same.

What you need

Sturdy cardboard / Two thin books / Large toy car

Use a smooth board with rounded edges instead of cardboard.

Before you begin

Place the ramp on the floor with one end on a book and a clear run-out area.

An adult secures a low ramp; nobody stands on it. Keep the car away from stairs.

Try it together

1. Release the car from the top without pushing and see where it stops.
2. Add the second book under the same end.
3. Release from the same mark and compare; repeat each slope once.

Talk, draw or show your idea

Did the car travel to the same place each time?

Space to explore

5 / 20 | Ages 3-4 | 15 minutes

From ice to water

What we are practising

Observe melting and describe solid and liquid forms of water.

What you need

Large piece of ice / Shallow unbreakable tray / Towel / Paper / Large crayons

Watch condensation droplets on the outside of a cold sealed bottle instead; explain they come from air.

Before you begin

An adult places the ice in the tray on a towel.

Adult supervision throughout; no tasting, prolonged ice contact or hot water. Empty the tray after use.

Try it together

1. Look at the ice and describe its shape without putting it in the mouth.
2. Leave it at room temperature while drawing what you see.
3. Check several times over ten minutes and describe any water that appears.

Talk, draw or show your idea

Where did the little puddle come from?

Space to explore

6 / 20 | Ages 3-4 | 5 minutes

Hearing the hidden bell

What we are practising

Link hearing with information from sounds.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a bell behind a screen and a child listening.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a bell behind a screen and a child listening.
2. Ask: The bell rings where we cannot see it. Which body part helps us hear it?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a bell behind a screen and a child listening. The bell rings where we cannot see it. Which body part helps us hear it?

Space to explore

7 / 20 | Ages 3-4 | 5 minutes

Real kitten, toy kitten

What we are practising

Distinguish an animal from its model.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a living kitten and a stuffed kitten.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a living kitten and a stuffed kitten.
2. Ask: Which needs food and water to stay alive?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a living kitten and a stuffed kitten. Which needs food and water to stay alive?

Space to explore

8 / 20 | Ages 3-4 | 5 minutes

The plant that needs a drink

What we are practising

Recognise a basic need of a living plant.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a living potted plant and a plastic plant.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a living potted plant and a plastic plant.
2. Ask: Which plant needs water to grow?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a living potted plant and a plastic plant. Which plant needs water to grow?

Space to explore

9 / 20 | Ages 3-4 | 5 minutes

Soft enough to squash

What we are practising

Compare materials by how they deform.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a soft sponge and a hard wooden block.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a soft sponge and a hard wooden block.
2. Ask: Which is easier to squash gently with a hand?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a soft sponge and a hard wooden block. Which is easier to squash gently with a hand?

Space to explore

10 / 20 | Ages 3-4 | 5 minutes

A roof against the rain

What we are practising

Follow a path to explain shelter.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw rain falling straight down onto a roof above a toy house.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw rain falling straight down onto a roof above a toy house.
2. Ask: What meets the falling drops first: the roof or the covered floor?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw rain falling straight down onto a roof above a toy house. What meets the falling drops first: the roof or the covered floor?

Space to explore

11 / 20 | Ages 3-4 | 5 minutes

Clouds are not cotton

What we are practising

Separate resemblance from material identity.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a fluffy cloud and a piece of cotton, clearly as two different things.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a fluffy cloud and a piece of cotton, clearly as two different things.
2. Ask: A cloud looks like cotton. Is it actually cotton?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a fluffy cloud and a piece of cotton, clearly as two different things. A cloud looks like cotton. Is it actually cotton?

Space to explore

12 / 20 | Ages 3-4 | 5 minutes

The cup without a drink

What we are practising

Recognise that unseen air occupies space.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw an open upright cup with no liquid or objects inside.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw an open upright cup with no liquid or objects inside.
2. Ask: Does this cup contain air even though we cannot see it?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw an open upright cup with no liquid or objects inside. Does this cup contain air even though we cannot see it?

Space to explore

13 / 20 | Ages 3-4 | 5 minutes

A lump becomes a pancake

What we are practising

Identify a change of shape.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a ball of modelling dough before and after it is pressed flat.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a ball of modelling dough before and after it is pressed flat.
2. Ask: Did pressing change its shape or turn it into a different material?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a ball of modelling dough before and after it is pressed flat. Did pressing change its shape or turn it into a different material?

Space to explore

14 / 20 | Ages 3-4 | 5 minutes

The bell that has stopped

What we are practising

Distinguish an object from its active sound-making state.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a ringing bell and then the same bell resting quietly.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a ringing bell and then the same bell resting quietly.
2. Ask: Must a bell make a sound all the time just because it is a bell?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a ringing bell and then the same bell resting quietly. Must a bell make a sound all the time just because it is a bell?

Space to explore

15 / 20 | Ages 3-4 | 5 minutes

What can a bean seed become?

What we are practising

Link a seed with a developing plant.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a bean seed and a stone beside pictures of a bean seedling and a toy car.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a bean seed and a stone beside pictures of a bean seedling and a toy car.
2. Ask: With suitable growing conditions, which could grow into the seedling: the seed or the stone?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a bean seed and a stone beside pictures of a bean seedling and a toy car. With suitable growing conditions, which could grow into the seedling: the seed or the stone?

Space to explore

16 / 20 | Ages 3-4 | 5 minutes

Which picture shows a push?

What we are practising

Distinguish a push from a pull.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a hand moving a toy away and another bringing a toy closer.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a hand moving a toy away and another bringing a toy closer.
2. Ask: Which hand pushes the toy away?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a hand moving a toy away and another bringing a toy closer. Which hand pushes the toy away?

Space to explore

17 / 20 | Ages 3-4 | 5 minutes

Rain is falling water

What we are practising

Identify rain as liquid water.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw rain drops outside a window.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw rain drops outside a window.
2. Ask: Are ordinary raindrops little pieces of paper or drops of water?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw rain drops outside a window. Are ordinary raindrops little pieces of paper or drops of water?

Space to explore

18 / 20 | Ages 3-4 | 5 minutes

The rabbit's outside covering

What we are practising

Compare visible animal coverings.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a rabbit with fur and a duck with feathers.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a rabbit with fur and a duck with feathers.
2. Ask: Which covering belongs to the rabbit: fur or feathers?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a rabbit with fur and a duck with feathers. Which covering belongs to the rabbit: fur or feathers?

Space to explore

19 / 20 | Ages 3-4 | 5 minutes

Wheels go round

What we are practising

Describe rotation using a marked point.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a toy car's round wheel with a dot near its edge, at two moments as the wheel turns.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a toy car's round wheel with a dot near its edge, at two moments as the wheel turns.
2. Ask: Does the marked dot stay at the top while the wheel turns?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a toy car's round wheel with a dot near its edge, at two moments as the wheel turns. Does the marked dot stay at the top while the wheel turns?

Space to explore

20 / 20 | Ages 3-4 | 5 minutes

See the stem's support

What we are practising

Identify a supporting plant structure.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a sunflower with roots below, a stem, leaves and a flower above.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a sunflower with roots below, a stem, leaves and a flower above.
2. Ask: Which part holds the flower above the ground: the stem or the petals?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a sunflower with roots below, a stem, leaves and a flower above. Which part holds the flower above the ground: the stem or the petals?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. Hand shadow shapes

A little hint

Compare how much wall the shadow covers.

Answer / what to notice

With the wall and torch fixed, moving the hand towards the torch usually makes its shadow larger.

The hand blocks some light; its position changes the blocked area on the wall.

Make it easier

Watch one adult-made shadow and copy its hand shape.

Take it further

Predict which position makes the smaller shadow, then test.

Another day, away from the screen

Trace an outdoor object's shadow with your finger.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Float or sink tray

A little hint

Look at whether the object stays at the surface or rests below it.

Answer / what to notice

Record what happens: a wooden spoon often floats and a metal spoon usually sinks; cup behaviour depends on its material and trapped air.

Material, shape and trapped air can all affect floating; size alone does not decide it.

Make it easier

Test just the two spoons.

Take it further

Test the cup again and explain what changed.

Another day, away from the screen

Notice floating objects in a picture book.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. Move it with air

A little hint

Feel the moving air on your own hand.

Answer / what to notice

Air moved by the fan can push the paper; results depend on distance and how strongly you fan.

Air is hard to see, but we can observe what moving air does.

Make it easier

An adult fans while the child watches and describes.

Take it further

Keep the distance fixed and compare gentle and stronger fanning.

Another day, away from the screen

Watch leaves move in the breeze from a safe spot.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. A gentle ramp test

A little hint

Use a hand beside the stopping point as a temporary marker.

Answer / what to notice

The higher ramp often sends the car farther, but bumps and friction can change the result.

A fair comparison uses the same car, surface and release point.

Make it easier

Use one slope and practise releasing without pushing.

Take it further

Try three releases on each slope and describe the pattern.

Another day, away from the screen

Look for real ramps used by wheels nearby.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. From ice to water

A little hint

Look at the shrinking edges of the ice.

Answer / what to notice

The ice melts into liquid water; a large piece may take longer than the activity to melt fully.

Warm surroundings transfer heat to the ice, helping it melt.

Make it easier

Simply point to ice and liquid water.

Take it further

Draw the ice at the beginning and after ten minutes.

Another day, away from the screen

Ask an adult to freeze water again and check it later.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

6. Hearing the hidden bell

A little hint

Think about listening, not looking.

Answer / what to notice

Our ears.

Hearing can give information even when the sound source is hidden.

Make it easier

Point to ears while making a quiet bell sound.

Take it further

Name something hearing cannot tell us, such as the hidden bell's colour.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

7. Real kitten, toy kitten

A little hint

A toy can look like an animal without being alive.

Answer / what to notice

The living kitten.

Living animals have needs that a stuffed toy does not have.

Make it easier

Label each drawing real or toy.

Take it further

Explain why moving a toy by hand does not make it alive.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

8. The plant that needs a drink

A little hint

Ask which one is living.

Answer / what to notice

The living plant.

Water supports living plants; a plastic model does not grow.

Make it easier

Point out soil and real leaves in the drawing.

Take it further

Name another need of most green plants: suitable light.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

9. Soft enough to squash

A little hint

Think about soft and hard.

Answer / what to notice

The soft sponge.

Its shape changes under a gentle squeeze more readily than the wooden block's.

Make it easier

Show a picture of the squeezed sponge.

Take it further

Explain why soft does not necessarily mean liquid.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

10. A roof against the rain

A little hint

Follow a drop's downward path.

Answer / what to notice

The roof.

In this drawing, the roof lies between the falling drops and the floor.

Make it easier

Trace one drop with a finger.

Take it further

Add a hole in the roof and predict where a drip might appear.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

11. Clouds are not cotton

A little hint

Things can look alike but be made of different materials.

Answer / what to notice

No. Clouds contain tiny water droplets or ice crystals.

Appearance alone does not tell us what something is made of.

Make it easier

Say “looks like” instead of “is”.

Take it further

Find another picture of things that look alike but differ, such as a toy apple and a real apple.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

12. The cup without a drink

A little hint

Air fills open spaces around us.

Answer / what to notice

Yes, it contains air.

“No drink inside” does not mean there is no air inside.

Make it easier

Compare the cup's inside with the air around it.

Take it further

Draw air as pretend dots and explain that the dots are a model, not what we normally see.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

13. A lump becomes a pancake

A little hint

It is still dough in both pictures.

Answer / what to notice

Its shape changed; it is still dough.

A change of shape does not by itself make a new material.

Make it easier

Name the ball and flat shapes.

Take it further

Suggest how to roll it back into a ball.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

14. The bell that has stopped

A little hint

A bell can be still and quiet.

Answer / what to notice

No. It makes sound when it vibrates enough to be heard.

A familiar sound source can be silent when its motion stops.

Make it easier

Mime ringing and stopping.

Take it further

Name another object that can be noisy or quiet.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

15. What can a bean seed become?

A little hint

A seed contains a young plant.

Answer / what to notice

The bean seed could.

The stone does not contain the growing plant found in a seed; seeds still need suitable conditions.

Make it easier

Connect the seed and seedling with a line.

Take it further

Explain why planting a toy car will not grow a bigger car.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

16. Which picture shows a push?

A little hint

A push acts away from the pushing hand.

Answer / what to notice

The hand moving the toy away.

The other drawing shows a pull towards the hand.

Make it easier

Act with imaginary toys while staying seated.

Take it further

Describe a drawer being opened and closed using pull and push.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

17. Rain is falling water

A little hint

Think about a puddle after rain.

Answer / what to notice

Drops of liquid water.

Rain can collect on the ground as water; it is not paper confetti.

Make it easier

Draw a puddle below the rain.

Take it further

Compare rain with frozen precipitation such as hail using pictures only.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

18. The rabbit's outside covering

A little hint

Compare the two pictured animals.

Answer / what to notice

Fur.

Rabbits are mammals with fur; ducks are birds with feathers.

Make it easier

Point to the rabbit while naming fur.

Take it further

Name another furry animal and another feathered bird.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

19. Wheels go round

A little hint

The dot is attached to the wheel.

Answer / what to notice

No; the dot moves around with the wheel.

Parts of a rotating wheel change position around its centre.

Make it easier

Turn a paper drawing slowly.

Take it further

Predict where a top dot goes after half a turn: bottom.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

20. See the stem's support

A little hint

Follow the tall supporting part.

Answer / what to notice

The stem.

The stem supports the above-ground parts; petals are parts of the flower.

Make it easier

Trace the stem from soil to flower.

Take it further

Add leaves and show where they join the stem.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.